



LOGIC DESIGN

FIRST EXAM

موعد المحاضرة: (1-3)

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Q1) Fill in table below the correct answer:

1- $(151)_{10} = (\dots\dots\dots)_2$	a- 1011	b- 1101	c- 10010111	d- 204
2- A binary representation of $(-34)_{10}$ when treated as a 8-bit signed magnitude number is	a- 11011101	b- 11100010	c- 10100010	d- 11011110
3- The binary result for the decimal $(-24) + (-15)$ using 8-bit signed 2's complement is	a- 10100111	b- 11100111	c- 11011001	d- Nothing above
4- How many gates would be required to implement the following Boolean expression before simplification? $XY = X(X+Z) + Y(X+Z)$	a- 1	b- 2	c- 4	d- 5
5- The complement function of $F = x(yz + yz)$ is: $\bar{x} + ((y+z)(\bar{y}+\bar{z})) = \bar{x} + y\bar{z} + z\bar{y}$	a- $x(\bar{y}\bar{z}) + (\bar{y}\bar{z})$	b- $\bar{x}(y+z) + (\bar{y}+\bar{z})$	c- $\bar{x} + y\bar{z} + \bar{y}z$	d- All of them
6- Excess-3 code is known as.....code :	a- Weighted	b- Cyclic	c- Self-complementing	d- Algebraic
7- $(192.123)_8 = (\dots\dots\dots)_{10}$	a- 300 173	b- 138.83	c- 138.0767	d- Nothing Above
8- To detect errors in data communication, a bit is sometimes added to ASCII code that is called:	a- Gray	b- Detector	c- Acknowledge	d- Parity
9- The function $F = x + (xy) = (x+z)(x+y)$	a- x	b- $x+y$	c- y	d- xy
10- $AC + ABC + AC(1+3) = A$	a- A	b- AC	c- AB	d- 1

1	2	3	4	5	6	7	8	9	10
b	c	c	d	c	c	c	d	b	b

Q2) Using Boolean Algebra, reduce the function $F = AB + \bar{A}\bar{C} + A\bar{B}C(AB + C)$
الاجابة هي التالي

Q3) Write the minimized function for $F_{1,2,6,7,8} = \sum(1,2,6,7,8)$, $d_{10,11,12,13,14,15} = \sum(10,11,12,13,14,15)$

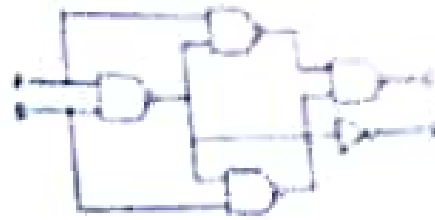
الحل $\rightarrow$

$$F(A,B,C,D) = \bar{A}\bar{B}\bar{C}D + \bar{C}\bar{D} + B\bar{C} + A\bar{B}\bar{D}$$

CD

AB	00	01	11	10
00	0	1	0	1
01	0	0	1	1
11	X	X	X	X
10	1	0	X	X

Q1) For the following circuit, Name the equivalent gates that output X and Y are representing, Show your solution.



Q2) Implement the circuit below using only NAND gates.



Q3) Design and build a five bits odd parity generator.

Q4) Design and build a 3x3 bits Multiplier.

Q5) Write the function for the circuit below.

